Schwann cells culture with adipose stem cells and neuregulin to determine influencing factors applicable in nerve regeneration

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Abstract

Schwann cell (SC) cultures are required for neural regeneration, as for example in injured nerves. However, SC culture procedures need improvements. This study evaluates and compares growth of SC cultures without and with neuregulin, and in co-cultures with adipose mesenchymal stem cells (ASCs). Rat originated SCs and ASCs were cultured, maintained and subcultured to prepare monocultures and co-cultures for this study. Morphological, immunohistochemical and quantitative evaluations were performed at 48 and 96 h. In monocultures SCs were S100 and vimentin positive, appeared bipolar with small somata containing small and elongated nuclei, and tended to contact on their ends forming chains or clustering around other cell types. ASCs were CD 105 and vimentin positive, showed fibroblast-like shape with large rounded nuclei and distinct nucleoli, became larger along time and contacted with the neighboring cells. In SC-ASC co-culture, a few ASCs appeared in triangular shape, their CD105 positivity turned weaker and became vimentin negative at 4 days confirming ASC differentiation initiation towards SC. Fibroblasts appeared in a small proportion in each culture. Cell counting and further ANOVA analysis revealed significant differences for cell growth between 48 and 96 h in each culture, whereas across different cultures, significant differences were found only between SC cultures grown without neuregulin and rest of SC cultures. In co-cultures ASC proliferated faster. Neuregulin increases SC cultures growth about 4 times faster than SC cultures alone. A similar SC proliferation increase was observed in co-cultures of SC with ASC at 96 h, suggesting possible role of ASC released growth factors in SC growth.

Keywords

Nerve Regeneration, Schwann, Adipose-derived mesenchymal stem cells, Neuregulin, Co-culture

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Introduction

Traumatic and degenerative diseases of the nervous system (NS) show high and rising incidence. For example, those diseases caused by trauma to peripheral nervous system (PNS) reach around 300,000 cases per year in Europe (Wiberg and Terenghi, 2003) and have quite high treatment costs and loss of work hours (Rosberg et al., 2005). Functional recovery is slow in peripheral and almost nonexistent at central nervous system. This recovery is quite poor in sensory aspect and reduced regarding motor functions, and shows that regeneration difficulties limit treatment effectiveness. On other prospect, consequences are often severe depending on amount of injured neurons, capacity for regeneration of nerve fiber and distance to proximal and distal end portions of injured nerve or injury site. Spontaneous repair of neurotmesis does not usually occur, making it necessary to bridge the gap by means of a graft. In this regard, the “gold standard” is considered the autologous graft, in which a defect in an injured motor nerve is substituted by a segment of an unimportant sensory nerve (Millesi et al., 1972). This repairing may be improved by the addition of growth factors, electrical stimulation, phototherapy and physiotherapy (Deumens et al., 2010; García Medrano et al., 2014). Since the obtention of a segment of autologous sensitive nerve to be grafted carry morbidity and side-effects, other omplants have been used such as predegenerated muscle (García Medrano et al., 2013), veins, tendons and non-autologous biological derivatives, as well as biocompatible synthetic materials (Garrosa García et al., 2011; Kalbermatten et al., 2008; Siemionow et al., 2010). Currently, there are interests seeking more new options, since it is clear that most meticulous surgical repair does not guarantee best functional recovery.

The key to possible repair seems to be in SCs, which play an active role in recovery process of peripheral fibers facilitating and stimulating axonal regeneration and providing structural support to guide regenerating nerve fiber (Mahay et al., 2008). At distal end of nerve section, SCs not only participate in phagocytoses of Wallerian degeneration remains, but they also undergo a dedifferentiation process to SC precursors and proliferate organizing Büngner’s bands, and produce extracellular matrix molecules, integrins, trophic factors that direct regenerating axons and later, myelin (Wiberg and Terenghi, 2003).

The important role of SC in neurogenesis, neuronal survival and axonal growth in injured nerves in PNS and at neuromuscular junction has been successfully demonstrated (Jessen and Mirsky, 2016). Due to these characteristics they have become focus of numerous investigations for exploring their possible therapeutic role in demyelinating, autoimmune, degenerative or traumatic diseases in NS as well as diseases affecting neuromuscular junction. Hence, isolation, acquisition, cultivation and purification of SCs are an interesting field of research and, above all, with many possibilities for therapeutic application (Vroemen and Weidner, 2003). Initially they have been harvested from embryonic or neonatal cell cultures but also from the adult, although these latter cells differ in expression of some receptors (Ansselin et al., 1997; de la Fuente et al., 2012). Hence, efficient cultivation methods are in great need to find information related to properties of their membranes on communication between them, and between them and axon; and also to its possible alteration by neuropathic conditions. The answers to these questions could facilitate repair of NS damage. Increase in growth number of SC to facilitate their use as a therapeutic method in NS injury is a challenge for regenerative medicine, and more critical when it is already recognized importance of its characteristics and functional properties for nervous tissue.

Currently, research interests are focused on use of neurotrophic factors released at injury site to increase regeneration of PNS, which could be applied to fulfill same function in CNS. Hence, their study and possibility to enhance growth potential is a challenge for research. In this process it has

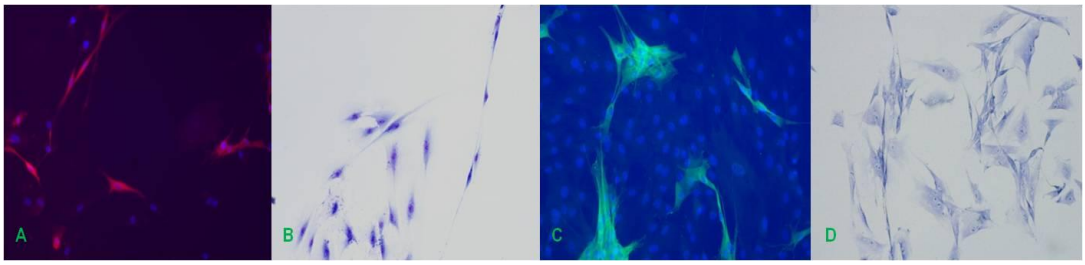


Figure 1 Schwann cells obtained from sciatic nerve of rat were cultivated without neuregulin for 48 (A and B) and 96 (C and D) hours and stained with anti-S100 (red or green; SC marker) and DAPI (blue; nucleus marker) (A and C) and Giemsa stain (violet; morphology) (B and D) and observed under microscope using 20x objective.

raised interest to use different types of conduits or grafts that can be placed at site of lesion, between two injured nerve terminals, and eventually containing in vitro cultured SC or stem cell differentiated SC or SC co-cultured with stem cells (Brohlin et al., 2009).

The neurons of PNS require SC and certain components of its basal lamina to stimulate their neurite growth. An appropriate substrate and a series of chemical substances, including growth factors or neurotrophic factors are also essential for this growth. Some of them have already expressed during development of NS. During nerve injury, it has been postulated production of a new profile of proteins including neurotrophic factors, which are substances that promote cell growth and appear to have defined roles in neuronal regeneration. The secretion of many neurotrophic factors is increased during nerve injury. The best known, and associated with changes in neuronal regeneration, neurotrophic factors are Nerve Growth Factor (NGF), Brain-Derived Neurotrophic Factor (BDNF), Ciliary Neurotrophic Factor (CNTF), Glial cells Derived Neurotrophic Factor (GDNF), Neurotrophins (NT) and Neuregulins (NRG).

Neuregulins are a family of soluble or membrane-bound proteins. Their expression in axon is necessary for differentiation, maturation and survival of immature SC. They are present in neuronal cells acting as signal mediators for autocrine / paracrine or juxtacrine secretion. They could be partly responsible for proliferation during Wallerian regeneration (Carroll et al., 1997; Raabe et al., 1996). Also, they play a key role in process of myelination and provide a guiding signal in organizing axonal cover (Taveggia et al., 2005), but still little is known about signaling mechanisms and their regulation. In addition, neuregulin could be used in therapy, but different concerns make its use inconvenient, such as the need of repeated injections due to its decrease with time in vivo.

Adult stem cells are present in developed organs and secrete growth factors, and may differentiate into different lines to provide cell turnover and replace cell loss in pathological situations. Some of these stem cells such as mesenchymal stem cells (MSC) and bone marrow-derived stem cells (BMSC) have been thoroughly used in regenerative medicine. Other stem cells are starting to be known and their applications appearing as promising possibilities. Adipose tissue-derived stem cells (ASC) are mesenchymal cells with ample similarities with BMSC but much easier to obtain and easily cultivated in vitro. Specifically, adipose tissue in inguinal region of rodents have shown a high degree of plasticity, and proliferative and differentiation capacity (Zhu et al., 2009). These cells may secret different growth factors, such as endothelial vascular growing factor (VEGF), hepatocyte growing factor (HGF), insulin-like growing factor (IGF-1), nerve growth factor (NGF), brain-derived neurotrophic factor (BDNF) granulocyte and monocyte clones stimulating factor (GM-CSF), transforming growth factor alpha (TGFα), as well as interleukines and adipokines among other factors. Thus, the secretion of different molecular profiles by ASCs can impose their effects at injury site (Alonso-Alonso and Srivastava, 2015; Rodriguez-Crespo et al., 2014; Singh et al., 2013), favoring SC proliferation, or else, transdifferentiation into SCs by means of appropriate stimuli and even transdifferentiation back into cells from every germ layer (Fu et al., 2016; Strem et al., 2005). These facts are bringing constantly new promises in cell therapy and regenerative medicine field. In addition, ASC shows less surgical complications for obtaining, as well as in most clinical trials, resulting and output in favor of patients.

This study evaluates SC growth and morphological characteristics in monoculture with and without neuregulin and in ASC and SC co-cultures, with an aim to increase yield of SC cultures for their possible use in cell therapy of nerve regeneration. In addition, the ASC growth and morphological characteristic in monoculture are also evaluated.

Materials and methods

Nerve predegeneration, isolation and culture of Schwann cells

The care and handling of animals was carried out according to guidelines of Council of European Communities (2010/636/EU) and Spanish State (RD 52/2013).

Forty adult Wistar rats of both sexes, weighing between 300 and 400 grams, aged 3 to 4 months, were taken for obtaining SC from sciatic nerve and MSC of adipose tissue. Each rat was anesthetized by intraperitoneal injection (1.5 ml / Kg body weight) of a mixture of 80% Ketamine and 20% Xylacine. Then, an incision of approximately two centimeters was performed in skin of thigh. Upon retraction of gluteal muscle, sciatic nerve was exposed and cut proximal to upper third of thigh and distally to next zone of bifurcation of tibia-fibula. The sectioned nerve was placed into a 3.5 cm culture plate (Falcon, ref 353001) containing a solution of Leibovitz with 1% antibiotic-antimycotic. In this plate material was cleaned by removing epineurium, traces of blood and other tissues, then transferred to another plate and chopped into portions of 2-5 mm in length. Surgery and cleaning were made under a microscope Optomic C 12.

Nerve portions were transferred into a Dulbecco modified medium (DMEM) with 10% FBS, 1% antibiotic-antimycotic and 10 nM neuregulin (8ul / ml) and placed in an incubator containing a culture condition of 37°C, 5% CO_2 and humidity saturation. In this condition they were allowed to undergo predegeneration for seven days. The seventh day they were removed from medium and placed in a tube (Falcon ref.352005), in a solution containing 2 ml of Leibovitz, antibiotic-antimycotic, 5% trypsin and 0.125% collagenase type II. In this solution they were left for three to four hours in a bath at 37 °C, shaking manually every 15 minutes. After this time DMEM (Gibco ref 31885-023) with 10% FBS and antibiotic-antimycotic was added and centrifuged at 2000 rpm for three minutes, then medium was removed. The pellet was resuspended in 1 ml of a solution of same medium. This solution was placed in two plates, one of which contained 2.5 ml of same medium with 10 nM neuregulin (8 ul / ml), and another plate did not contain neuregulin. The plates were transferred to an incubator at 37°C, 5% CO2

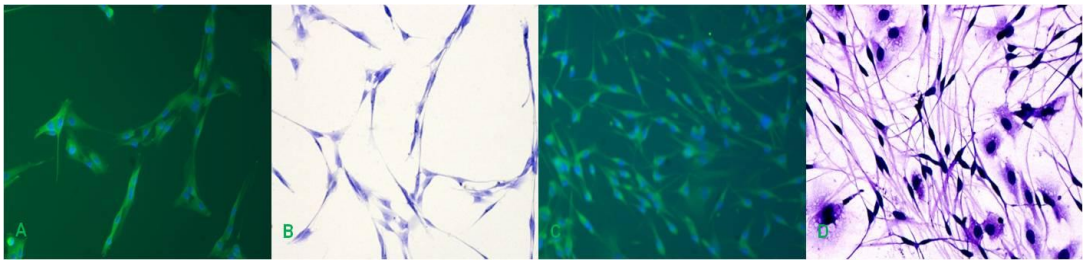


Figure 2 Schwann cells obtained from sciatic nerve of rat. These cells were cultivated with neuregulin for 48 (A and B) and 96 (C and D) hours and stained with anti-S100 (green; SC marker) and DAPI (blue; nucleus marker) (A and C) and Giemsa stain (violet; morphology) (B and D) and observed under microscope using 20x objective.

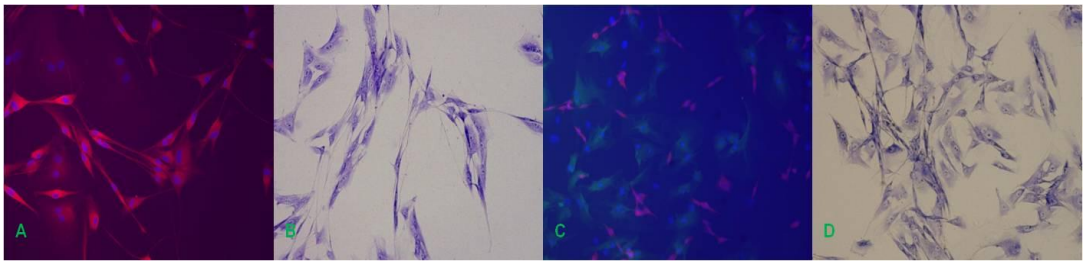


Figure 3 Co-culture of Schwann cells with adipose derived mesenchymal cells. The cells were stained at 48 hours with anti-vimentin (A, red; SC and ASC marker), anti CD 105 (C, green; ASC marker) and with anti S100 (C, red; SC marker). They were also stained with Giemsa (Violet; morphology) at 48 (B) and 96 (D) hours and DAPI (blue; nucleus). The cells were observed under microscope using objectives; 20x (A, B and D), and 10x (C).

and saturated humidity. When they reached confluence, they were reseeded on 4 plates; two without neuregulin and two with neuregulin.

Obtaining adult stem cells from adipose tissue

At eight days after sciatic nerve segment extraction, rats were anesthetized again and underwent an incision of about two centimeters in skin of inguinal region. One gram of adipose tissue was carefully dissected, removed aseptically and transferred to a plate containing culture medium of Leibovitz with 1% antibiotic-antimycotic. Traces of blood and other tissues were removed; adipose tissue was chopped into small pieces and then transferred to a new plate with same medium. Further it was incubated for enzymatic digestion for 60 minutes at 37°C in a solution containing 5 ml of Leibovitz and 0.075% collagenase type I. This mixture was passed through a filter of 60 um for separating undissociated tissue. Then collagenase effect is neutralized by addition of DMEM containing 10% Fetal Bovine Serum (FBS) and 1% antibiotic. The cell suspension was centrifuged at 1200 rpm for 10 minutes. The pellet was resuspended in a solution of ammonium chloride and left ten minutes in this medium and then centrifuged another 10 minutes at 1200 rpm. The new pellet is resuspended in DMEM at a density of 100,000 cells per ml and plated on three plates of 35 mm diameter, 1 ml per dish completing the volume up to 3 ml with same medium. The plates are incubated for thirty minutes in an incubator at 37°C with 5% CO_2 and saturated humidity. After this time it was taken out from incubator and medium was removed to discard non-adherent cells. The 3 ml of DMEM with 1% antibiotic and 10% FBS was added to each plate and they were maintained in same conditions in incubator. Culture medium was changed every two days up to reaching confluence.

Co-Culture of Schwann cells and adult stem cells from adipose tissue

It has been used ASC of 1 to 5 subcultures and SC of 1 to 2 subcultures for co-culture studies. When SC and ASC approached to become confluent occupying at least 80% of culture plate area, medium was removed and cells were washed with Hank's solution. Then a solution of Trypsin-Hank was added in plates containing ASC and trypsin solution in plates containing SC. The solutions were removed when it was found that cells were detached (approximately one minute incubation).

The DMEM with antibiotics and FBS were added in each plate to neutralize enzyme effect, and then it was washed by centrifugation. The cells were counted using trypan blue solution and seeded in ratio of 100,000 cells per ml. The volume of 1 ml per dish was completed with two milliliters more of DMEM. Similar number of SC and ASC were seed in each plate for co-cultures. The co-cultures were organized in groups of eight plates, two of which contained a mixture of SC and ASC, two contained SC seeded without neuregulin, two contained SC seeded with neuregulin and another two seeded with only ASC. The culture plates were transferred to incubator under the aforementioned conditions. At 48 and 96 h, immunohistochemical or Giemsa stains were performed.

Immunohistochemistry and cell counting

For immunohistochemical studies plates were washed with PBS at 48 and 96 h and fixed with 4% paraformaldehyde for 30 minutes at room temperature. The paraformaldehyde was removed, and plates were washed again three times with PBS (five minutes each time). Then cells were permeabilized with 0.2% Triton X-100, (except when the CD 105 antibodies are used), and were immunoblocked with 0.5% goat serum.

The primary antibodies were polyclonal anti-S100, anti-CD 105 and anti-vimentin diluted in PBS. The anti-S100 is considered characteristic of SC and was added to plates with cells taken from sciatic nerve and to plates containing co-cultures. The anti-CD 105 antibody was used to identify ASC and anti-vimentin for two cell types.

Cells were incubated with each primary antibody for 24 hours at 4 °C, and then it was removed. Again cells were washed with PBS three times (five minutes each time), and secondary antibody was applied. The latter was left in darkness at 4 °C for two hours, and at the end of which secondary antibody was removed. It was washed again with PBS three times (five minutes each time). Following, DAPI was used to stain nuclei, by allowing to act on cells for 10 minutes in dark. It was removed and washing was performed again three times with PBS. At 48 and 96 h photographs of three at randomly selected areas in each plate were taken using fluorescence microscopy with different combinations of filters for blue, green and red fluorescence under 10x and 20x objectives. These photographs were used to analyse cells with ASC phenotype and SC phenotype. Another set of plates was used for staining with Giemsa. They were subsequently taken to light microscope where they were counted. The counting was performed on screen showing microscope image. The surfaces of 35-mm plates were divided into 49 fields, which are marked with letters and numbers to avoid repetition, and 12 fields of each plate randomly selected were counted. The SC and ASC were identified by morphology and those cells that did not show these morphologies were considered as other cells. Data obtained were subjected to statistical analysis by SPSS program using one-way ANOVA and "post hoc" tests of Dunnet, Bonferroni and Tukey.

Results

Schwann cells cultured without neuregulin

In the plates in which SCs were seeded with media without neuregulin containing 10% FBS, it was observed that different types of cells grew slowly, covering the plate from 10% to 40% at 48 h, and between 10% to 50% at 96 h. It was obtained a mixture of cells in which it was noted that between 10% and 50% cells showed elongated and oval shaped soma with a similar shaped nucleus and two long bipolar extensions, which tended to contact with extensions of other cells. Such cells were defined as cells with schwanoide aspect and they were positive for S100 and vimentin proteins staining (Fig. 1 A), thus were classified as SC. Other cells were not positive for S100 protein expression; nuclei stained with DAPI were round, and generally showed varying morphology with fibroblastic appearance, and therefore classified as fibroblasts. SCs tended to cluster on or around fibroblasts, and grew slowly, whereas number of fibroblasts increased rapidly. After 96 h many of these fibroblasts were observed in some plates, while number of SCs increased slowly or moderately in some plates and in others tended to decrease (Fig. 1 B, C and D).

Schwann cells with neuregulin

In the plates where SCs were seeded with media (containing 10% FBS) supplemented with neuregulin, a rapid proliferation of bipolar cells with soma and small and elongated nuclei was observed, which tended to make contacts at ends forming chains or clustering around other cell types at 48 h. These cells showed same morphological characteristics as described for SC grown without neuregulin. They were positive for anti-S100 protein confirming to be SC. However, their proliferation rate was higher. It was also noted that number of cells showing schwannoid aspect exceeded number of cells showing fibroblast appearance by 9 folds, on the contrary to what happened with cells grown without neuregulin, where fibroblast appearance cells are more numerous. They showed positivity to vimentin protein (Fig. 2 A and B) and negative to S100 protein as had been seen in plates grown without neuregulin (Fig. 2 A). Culture plate was covered by cells between 40-60 %.

At 96 h culture plate was covered by cells between 70% and 100%. Most of these cells showed schwanoid appearance and were positive for S100 protein. It was also observed very few cells of fibroblast-like aspect (15% to 5%), which were not positive for S100 (Fig. 2 B and C). SCs made contacts among themselves and with fibroblasts and tended to grow over other cells. The purity of these cultures for SC ranged from 85% to 95%. The remaining cells, mostly showing fibroblastic appearance (5% to 15%), were larger than bipolar cells, contained only short cytoplasmic extensions, were positive for anti-vimentin antibodies, and showed large, oval or round nuclei stained with DAPI (Fig. 2 D). At 48 h percentage of plate surface covered with SC cultured with neuregulin was 40% to 60% (Fig. 2 B and C), which was already described, increases up to between 85% to 95% at 96 h (Table 1) (Fig. 1 A and B).

Tabla 1 Percentage of plate covered and numbers of SC and ASC per microscopic field in different culture and co-culture conditions at different time intervals.

Culture type	Variables	% Surface	SC/Field	ASC/Field
Nerve	Without NRG/48h	14885	10.56±3.21	
	Without NRG/96h	35-50	19.56±2.75	
	With NRG/48h	40-60	41±4.98	
	With NRG/96h	85-95	76±5.81	
Adipose Tissue	48h			43.39±4.06
	96h			77±4.80
Co-culture	48h	40-60	40.50±3.88	42.33±4.86
	96h	80-100	72.61±5.45	72.44±4.42

Schwann cells with adipose MSC in co-culture

In co-culture plates with SC and ASC at 48 h, it was observed that several types of cells including elongated, small spindle-shaped somata with elongated nuclei, in bipolar form with long processes, by means of which they were in contact forming chains. About 42% of cells stained for S100 and were classified as SC. The other types of cells were larger with round or oval nuclei and cytoplasm extended seeking contact with other cells. These cells were positive to CD105 and vimentin, so were classified as ASC. Besides these two predominant cell types it was observed some cells in triangular shape with three or more extensions leaving cell body and contacting to bipolar cells. These cells were positive to S100, so were classified as SC. A fourth cell type were distinguished by their polyhedral morphology, medium or large size and showed no positivity to S100 nor CD105. These cells were classified as fibroblasts. As time passed, SC showed accelerated growth, retaining their morphology, tendency to cluster with ASC or other cells, extending some prolongations that contacted with other SC and ASC. Both SC as ASC were vimentin positive at 48 h (Fig. 3).

After 48 h it was observed that cells had covered 40% to 60% of surface of culture plate which, at 96 h, reached between 80% and 100%. The SC retained their fusiform appearance; their processes extended and tended to make contact with each other at their ends, and also were grouped with ASC and tripolar cells. Among ASC it was observed numerous cells in mitosis. The CD105 expression in co-cultures was more intense at 48 h, while at 96 h it was weak. The SC showed no change regarding S100 protein expression at both 48 and 96 h and between cultures or co-cultures. The ASC were positive to Vimentin at 48 h, but not at 96 h in co-cultures. Morphologically, at 96 h there were the same four types of cells like at 48 h, i.e. elongated bipolar Scs, elongated tripolar SCs, ASCs and fibroblasts.(Fig 3 A, B, C and D).

The data related to total cell number per plate is presented in graph 1 which demonstrates increase and decrease tendency of cell number per plate at each time period for each type of cells growing in all these above mentioned conditions.

It has been added neuregulin to the co-culture only in a pilot study, but it has been not considered this group in the present study since the goal of this study was to evaluate the possibility to substitute neuregulin by ASCs in SC cultures, with the aim to implant these cells in future applications for nerve repair. The study offered evidence to consider that SC in culture grows in a comparable manner for both; with neuregulin alone and co-cultured with ASC.

Quantification of cell proliferation

At 48 and 96 h of culture, cells which were showing highest proliferation were ASC as confirmed by measuring number of cells per photographed field. At 48 h average density for ASC in cultures was 43.39 + 4.06, followed by ASC in co-cultures which showed a cell density of 42.33 +4.86. At third position SC cultures with neuregulin (41 + 4.98) and fourth was SC in co-cultures (40.5 + 3.88). At 96 h, average density of ASC in cultures were 77 + 4.8, followed by SC with neuregulin (76 + 5.81), SC in co-cultures (72.61 + 5.45) and ASC in co-cultures (72.44 + 4.42). The cultures which showed less proliferation was SC cultures without neuregulin (10.56 + 3.12; at 48 h, 19.56 + 2.75; at 96 h). This latter culture was only one which showed statistically significant differences with other four ($P < 0.001$). The SC cultivated with neuregulin and co-cultured with ASC showed similar behavior, whereas those grown without neuregulin proliferate significantly less ($P < 0.001$). The proliferation of ASC was similar in both cultures and co-cultures. A summary of these results are presented in table 1.

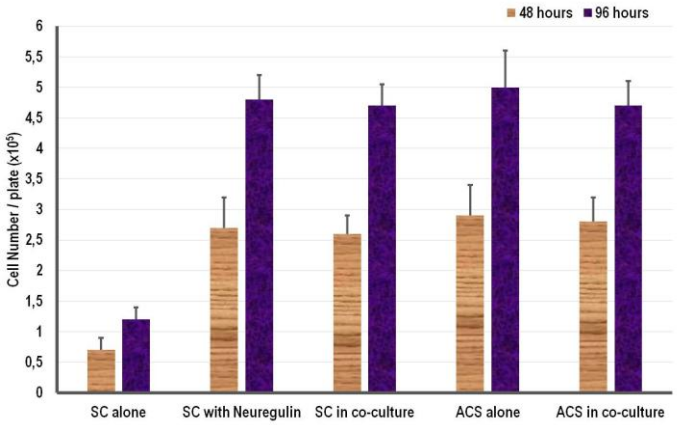
Discussion

Differentiation, maturation and survival of immature SC require a series of chemical substances such as neuregulins. In current scenario, it is well demonstrated that ASCs release various chemical substances, which are bioactive and could be obtained and then injected easily at injury site. Considering above opportunities for nerve regeneration, it becomes important to recognize those factors which can influence SC proliferation and this study was performed to reach a conclusion in this direction.

In this study adult animals were used as donors of both SC and ASC because they are easier to obtain and support to avoid ethical and carcinogenic problems concerning the use of embryonarian stem cells. The yield of SC obtained in this study was considerably high and with a purity of 90%. Perhaps it could be obtained higher yields by using newborn animals since these have a larger proliferative rate. But it was preferred to study the yield in adults taking into account its clinical application, since most of accidents and lesions occur in adult.

The results showed that in SC cultures with neuregulin and co-cultures with ASC, SC showed increased proliferation in comparison to cultures without neuregulin. This enhanced proliferation was statistically significant at both 48 and 96 h of culture. In SC cultures with neuregulin and co-cultures with ASC, SC showed a similar proliferation both at 48 and 96 h of cultures. The yield obtained in our culture of SC with neuregulin is high. Similar results have been reported by other authors, but using more complex methods, such as that by Brockes et al. (1979) (Brockes et al., 1979), who employed cytosine arabinoside, that by Kreider et al. (1982) (Kreider et al., 1982) who added poly-L-lysine for differential adhesion, or the one by Vroemer and Weidner (2003) (Vroemen and Weidner, 2003) using magnetic-activated cell separation. Regarding ASCs, they also show similar proliferation rate cultured alone and in the co-culture both at 48 and 96 h, while their size increases with time in cultures and co-cultures. This growth means population is doubled after 62.4 h, a similar time to that reported by Zuk et al. (2001) and Izadpanah et al. (2006) (Izadpanah et al., 2006; Zuk et al., 2001), who observed ASC population doubled after approximately 60 h.

Further calculations of data showed that main factors influencing to growth of cultures of SC and ASC are neuregulin (on SC), culture time (on SC and ASC), and co-culture (on SC and ASC). Culturing SC with neuregulin facilitated SC to multiply their numbers in cultures of 48 h by a factor of 3.88 and cultures of 96 h by a factor equal to 3.9. Thus, it was found that neuregulin produces a constant increase in cell number of SC, independent of cultivation time, and therefore confirming it is an influencing factor. In SC cultures without neuregulin, from 48 to 96 h, SC cell number is multiplied by 1.86, almost same (1.85) when neuregulin is supplied to cultures. The ASC in cultures is multiplied by a factor of 1.77, while in co-cultures it is somewhat lower, 1.71. Since there is a change from 48 to 92 h, it can be considered time as an influencing factor. Comparing SC cultures with neuregulin and co-cultures with ASC, it is found that SC numbers multiply by same in both cases, thus confirming similar effects for proliferation. It is also found that ASC multiplication factor is practically 1 i.e. these cells, like SC, multiply at same speed in cultures and co-cultures.



Graph 1 Diagram representing the number of cells obtained under the different experimental conditions at 48 and 96 h. There is significant statistical difference (P<0.01) for every experimental condition when compared 48 and 96 h of the same culture, while across the different cultures, statistical difference was only found in SC alone culture when compared with other cultures at 48 h (P<0.01) (*) and when compared at 96 h (P<0.01) (**).

The question arises whether increase in number of SC in co-culture is due only to SC proliferation or also due to transdifferentiation of ASC to SC. These both factors have been described for BMSC in 12 day-cultures with addition of PDGF, bFFG and neuregulin (Kingham et al., 2007; Tang et al., 2009; Wang et al., 2009). Our results seem to support that SC proliferation appears as the main factor since number of ASC cultured alone at 96 h is similar to that found in co-culture at same time. Nevertheless, phenotype change observed in ASC at 96 h of culture consisting in a diminution of positivity to CD-105 and turning vimentin negative makes think that they have already initiated their transdifferentiation (Zurita et al., 2005). These latter authors also have reported that BMSC co-cultured with SC, after one week, undergo transdifferentiation to a neural phenotype showing characteristic markers of SC, what is supported by "in vivo" studies (Chen et al., 2006; Vaquero and Zurita, 2009), and therefore this transdiffrentiation takes longer time than that covered by our study.

Thus, this study is able to provide data to evaluate and compare effects of ASC and SC co-culture and neuregulins to proliferation and morphological characteristics of SC and to determine influencing factors in this process. This could be applicable in planning a therapeutical option for nerve regeneration.

Acknowledgments

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